Low-Level Design (LLD) Specification

CredVault — Credit Card Management Platform

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Revision History

Version	Date	Author	Changes
0.1	2026-04-10	Engineering	Initial draft — service boundaries & DB schema
0.2	2026-04-12	Engineering	Added saga sequence diagrams & event catalog
0.3	2026-04-12	Engineering	DB migration: `AddMissingForeignKeys` applied to billing
1.0	2026-04-13	Engineering	Full LLD baseline — all sections complete

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1. Executive Summary & Design Principles

CredVault is a microservices-based credit card management platform built with ASP.NET Core 10, Angular 19, SQL Server, RabbitMQ, and Docker. The system allows users to register credit cards, track transactions, receive monthly bills, and pay them through a distributed, 2FA-secured payment flow.

1.1 Core Design Principles

Principle	Decision	Rationale
Database per Service	Each microservice owns an isolated SQL Server database	Enforces true service independence; prevents shared schema coupling
No Synchronous HTTP Between Services	All cross-service communication is event-driven via RabbitMQ	Eliminates temporal coupling and single points of failure
Clean Architecture	`Domain → Application → Infrastructure → API` layer separation per service	Business logic is framework-agnostic and fully testable
CQRS via MediatR	All writes use `IRequest<T>` Commands; all reads use Queries	Clear separation of read/write paths; independent scaling
Saga Orchestration	Payment flow uses MassTransit StateMachine saga	Resolves dual-write problem across 3 independent databases atomically
Eventual Consistency	Domain events trigger downstream reactions asynchronously	Accepts a short inconsistency window in exchange for high availability
Soft Deletes	Card service marks records `IsDeleted = true` instead of physical deletion	Maintains data history; prevents orphan foreign key issues
GUID Primary Keys	All entities use `uniqueidentifier` PKs	Enables ID generation at the application layer without DB round-trips

1.2 Technology Stack

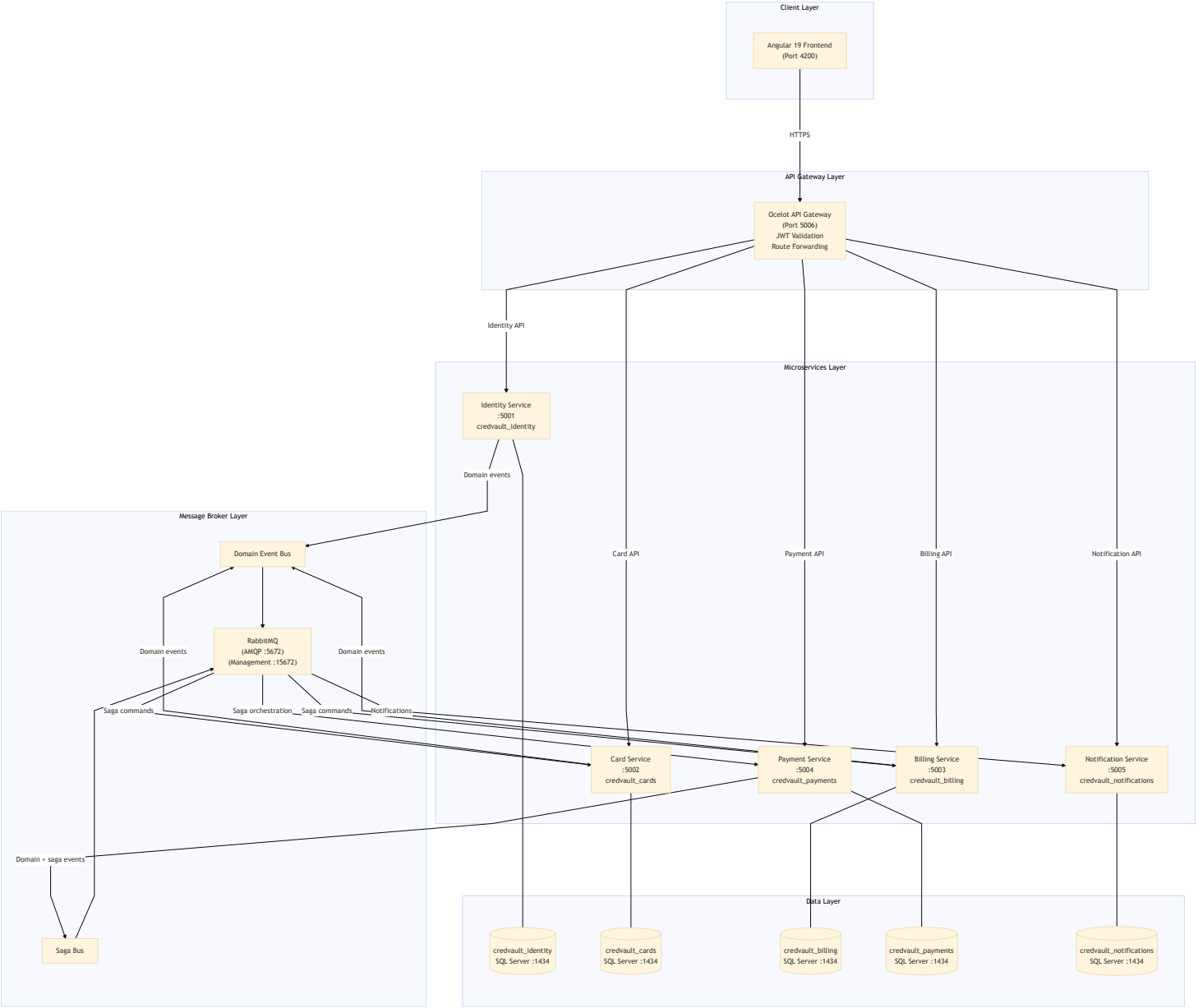
Layer	Technology
Backend Framework	ASP.NET Core 10 (.NET 10)
Frontend	Angular 19
ORM	Entity Framework Core 10
Database	Microsoft SQL Server (port 1434)
Message Broker	RabbitMQ with MassTransit
API Gateway	Ocelot
CQRS/Mediator	MediatR
Validation	FluentValidation
Email	SMTP / SendGrid
Containerization	Docker / Docker Compose
Auth	JWT Bearer tokens + OTP (6-digit, crypto-RNG)

2. Glossary of Terms

Term	Definition
Bounded Context	A service and its exclusively owned database/domain model
CQRS	Command Query Responsibility Segregation — writes (commands) and reads (queries) are separate pipelines
Saga	A long-lived transaction coordinator that manages a multi-step distributed workflow via events
MassTransit	.NET library abstracting RabbitMQ with built-in saga, retry, and consumer support
Domain Event	A fire-and-forget notification broadcast when something meaningful happens in a service
Saga Event	A request/response event within the payment orchestration; always consumed by the saga
DLQ	Dead Letter Queue — a RabbitMQ queue where failed/unprocessable messages land after retries are exhausted
OTP	One-Time Password — 6-digit code used for email verification and payment authorization
Compensation	Rolling back previous saga steps when a later step fails
Cross-service Reference	A GUID stored in Service B that logically refers to an entity owned by Service A, but with no SQL FK
Idempotent Consumer	A message consumer safe to call multiple times without side effects
Soft Delete	Marking `IsDeleted = true` rather than physically removing a row

3. System Architecture Overview

This diagram shows the full system from the client through to databases and the message broker.



Gateway Route Groups (for readability): `/api/v1/identity/**` , `/api/v1/cards/**` , `/api/v1/billing/**` , `/api/v1/payments/**` , `/api/v1/notifications/**`

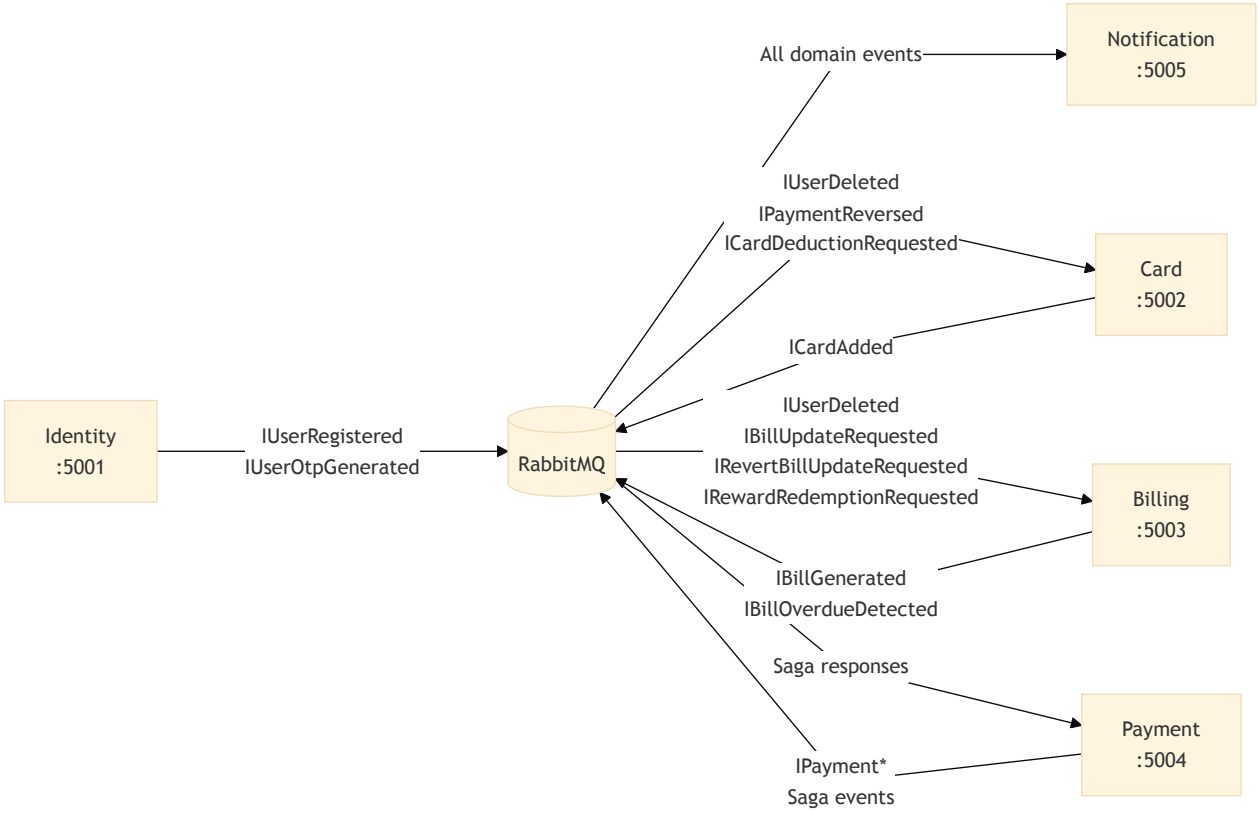
4. Bounded Contexts & Service Map

Each service is a fully independent deployable unit. The table below summarises ownership.

Service	Port	Database	Owns	Publishes	Consumes
Identity	5001	`credvault_identity`	Users, Auth	`IUserRegistered` , `IUserOtpGenerated`	*(none)*
Card	5002	`credvault_cards`	Cards, Issuers, Transactions	`ICardAdded` , `ICardDeductionSucceeded/Failed`	`IUserDeleted` , `IPaymentReversed` , `ICardDeductionRequested`
Billing	5003	`credvault_billing`	Bills, Statements, Rewards	`IBillGenerated` , `IBillOverdueDetected` , `IBillUpdateSucceeded/Failed`	`IUserDeleted` , `IBillUpdateRequested` , `IRevertBillUpdateRequested` , `IRewardRedemptionRequested`

Service	Port	Database	Owns	Publishes	Consumes
Payment	5004	`credvault_payments`	Payments, Ledger, Sagas	`IPaymentCompleted` , `IPaymentFailed` , `IPaymentOtpGenerated` , `IPaymentReversed` + all saga events	Saga orchestrates everything
Notification	5005	`credvault_notifications`	AuditLogs, NotificationLogs	*(none)*	All domain events — sends emails via SMTP
Gateway	5006	*(none)*	Routing, JWT enforcement	*(none)*	*(none)*

Service Dependency Graph

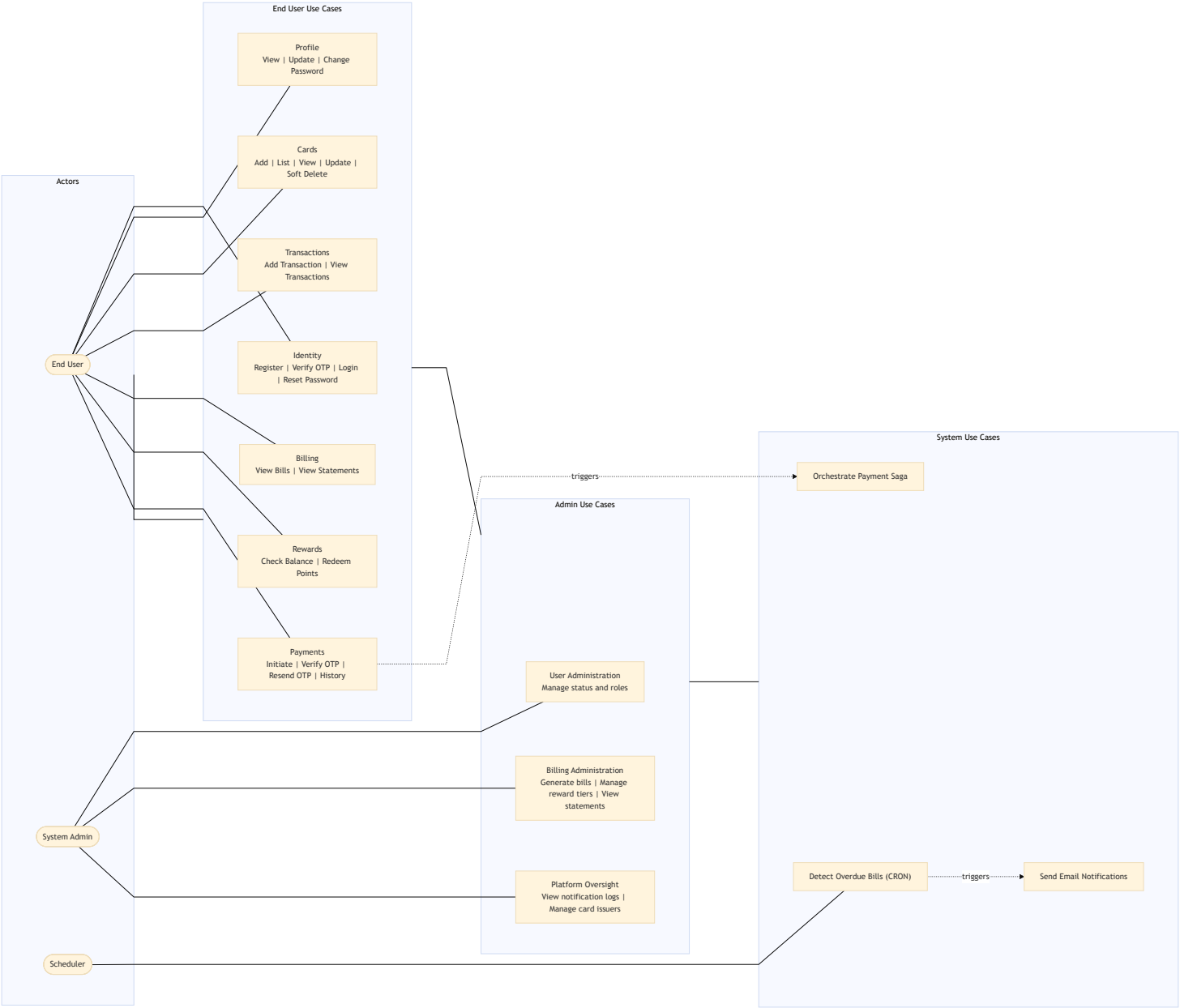


5. Actor & Use Case Model

5.1 System Actors

Actor	Description
End User	Registered customer who manages their own credit cards and payments
System Admin	Privileged operator with elevated access to manage users, cards, tiers, and billing
Scheduler	An internal background CRON timer that triggers overdue bill detection
Notification Worker	Internal async consumer that fires emails on domain events

5.2 Use Case Diagram



6. Class & Component Diagrams

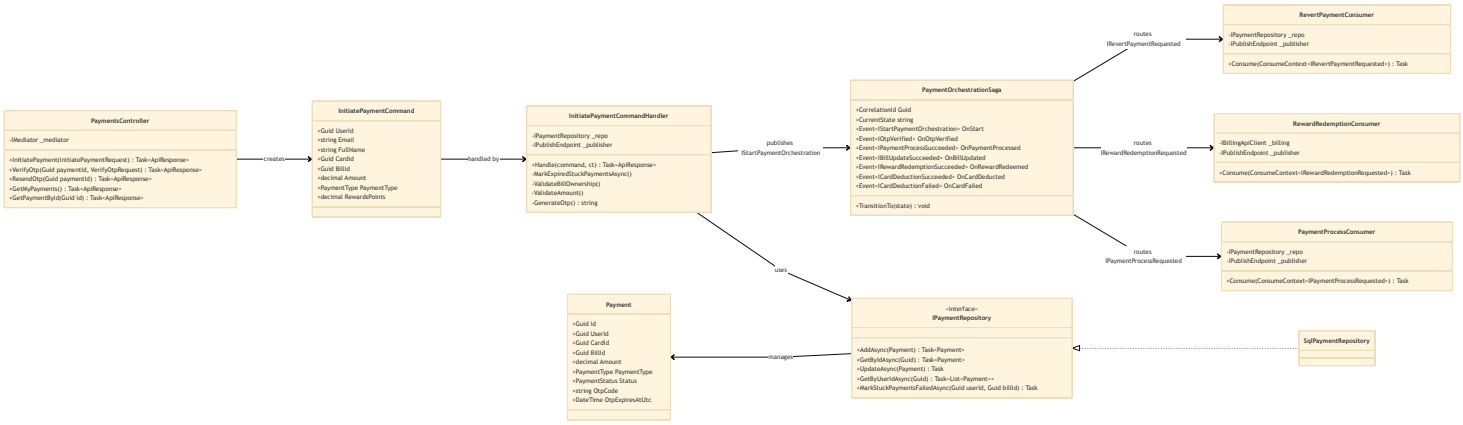
6.1 Clean Architecture Layers (Per Service)

Every service follows an identical 4-layer structure enforced at the project level.

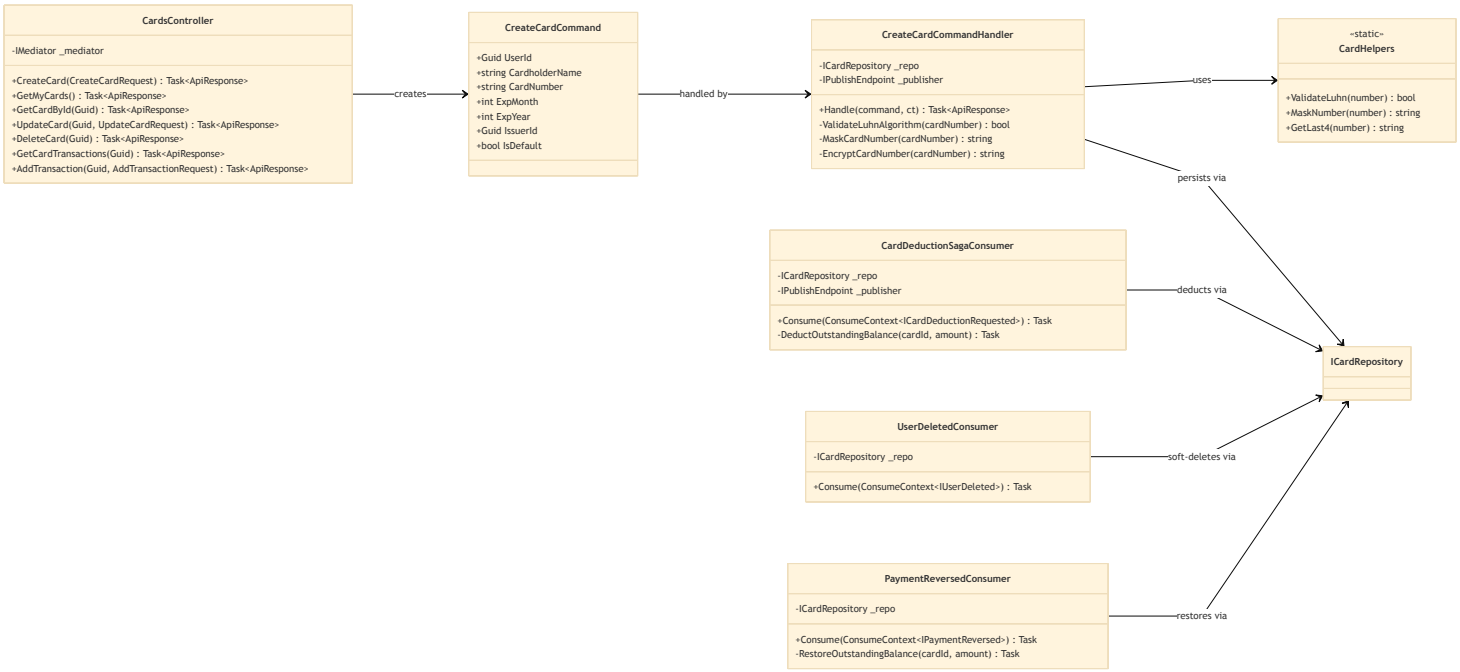
```
<ServiceName>/
├── <ServiceName>.Domain/
│   ├── Entities/         ← Core domain objects (no framework dependency)
│   └── Enums/            ← Status codes, types
│
├── <ServiceName>.Application/
│   ├── Commands/         ← Write-side: IRequest<T> + Handler
│   ├── Queries/          ← Read-side: IRequest<T> + Handler
│   ├── Interfaces/       ← Repository contracts (ICardRepository, etc.)
│   ├── DTOs/             ← Data Transfer Objects for API surface
│   └── Helpers/          ← Domain-specific utilities
│
├── <ServiceName>.Infrastructure/
│   ├── Persistence/Sql/  ← EF Core DbContext, Migrations, SqlRepositories
│   ├── Consumers/        ← MassTransit IConsumer<T> implementations
│   └── Services/         ← Email, external integrations
│
└── <ServiceName>.API/
    └── Controllers/      ← ASP.NET Controllers (thin; delegate to MediatR)
```

6.2 Payment Service — Full Class Diagram

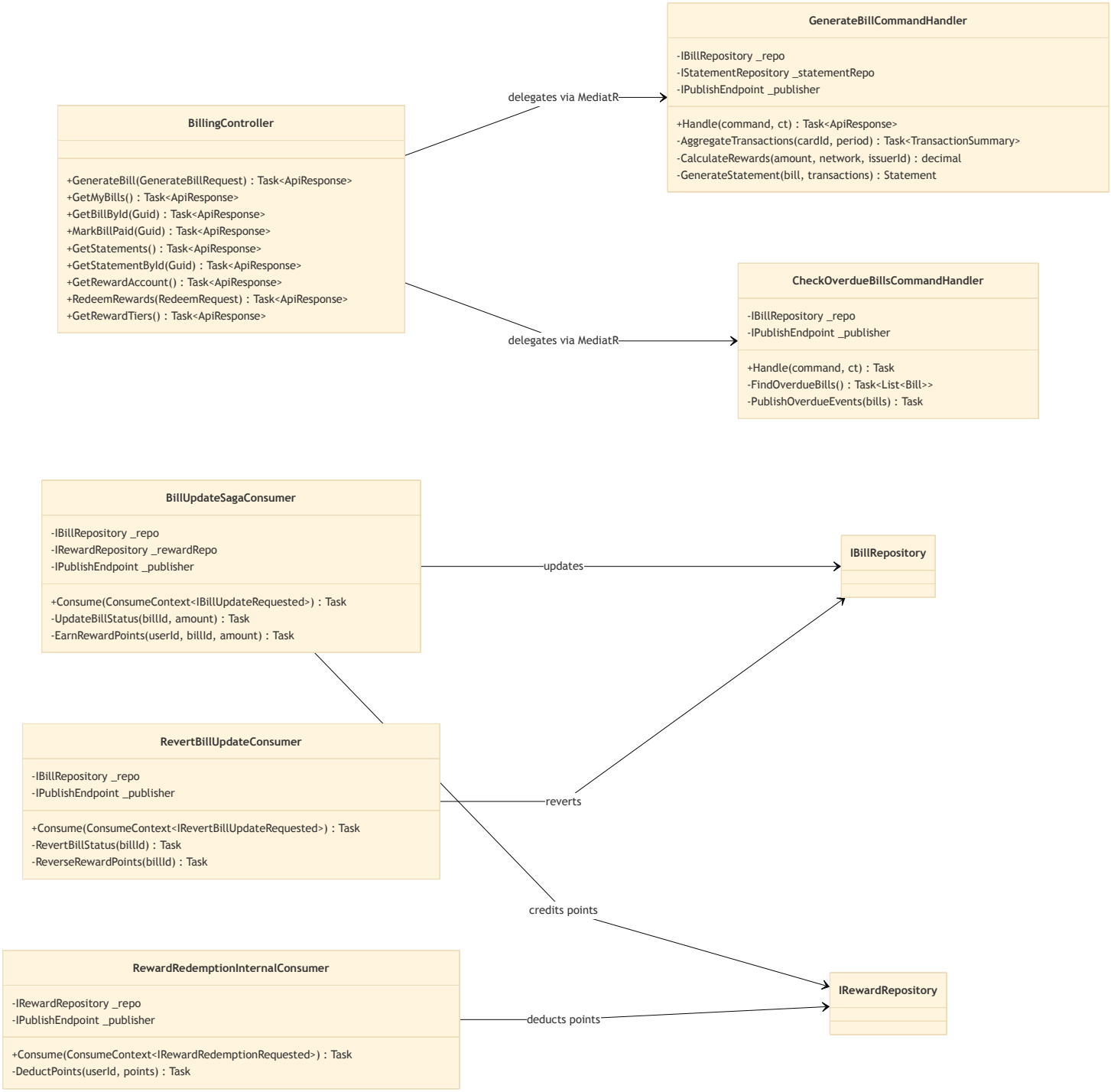
The Payment Service contains the highest complexity due to saga orchestration.



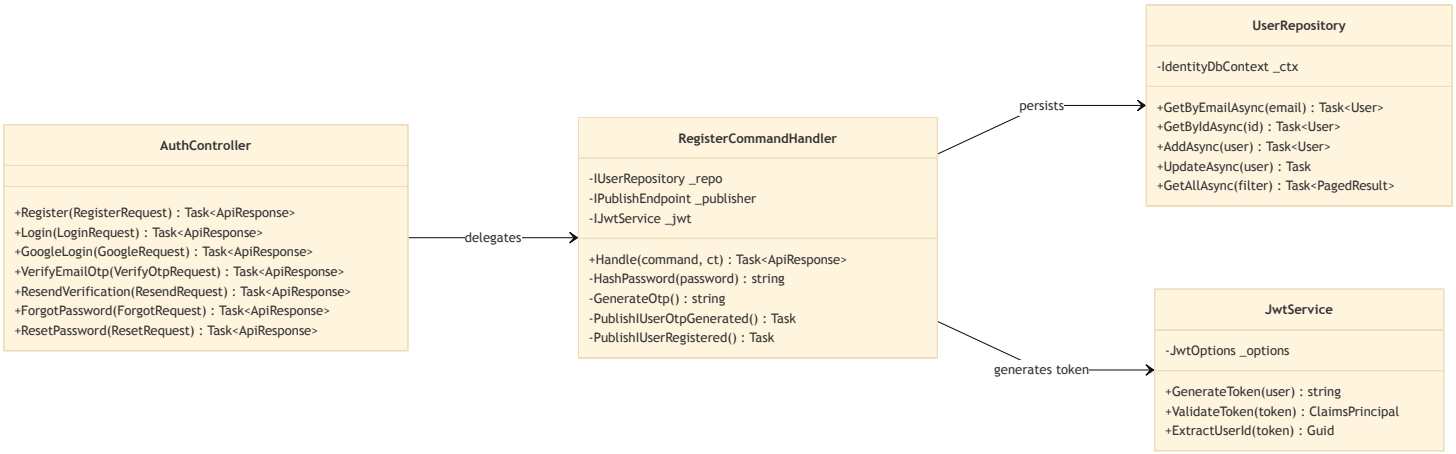
6.3 Card Service — Class Diagram



6.4 Billing Service — Class Diagram



6.5 Identity Service — Class Diagram



7. Database Architecture & ER Diagrams

All services follow the **Database-per-Service** pattern. SQL foreign keys only exist **within** the same database. Cross-service references are plain GUID columns with application-layer validation only.

7.1 Database Inventory

Service	Database	Tables	SQL FK Count	Last Migration
Identity	`credvault_identity`	1	0	Initial
Card	`credvault_cards`	3	2	Initial
Billing	`credvault_billing`	6	5	`AddMissingForeignKeys` (2026-04-12)
Payment	`credvault_payments`	3	1	Initial
Notification	`credvault_notifications`	2	0	Initial

7.2 Identity Database (`credvault_identity`)

identity_users			
uniqueidentifier	Id	PK	
string	Email		UNIQUE NOT NULL (256)
string	FullName		NOT NULL (256)
string	PasswordHash		NULLABLE (500)
bool	IsEmailVerified		
string	EmailVerificationOtp		(16)
datetime	EmailVerificationOtpExpiresAtUtc		
string	PasswordResetOtp		(16)
datetime	PasswordResetOtpExpiresAtUtc		
string	Status		pending-verification | active | suspended | blocked | deleted
string	Role		user | admin
datetime	CreatedAtUtc		
datetime	UpdatedAtUtc		

Indexes: `Email (UNIQUE)`

Enums:

Enum	Values
`UserStatus`	PendingVerification(1), Active(2), Suspended(3), Blocked(4), Deleted(5)
`UserRole`	User(1), Admin(2)

7.3 Card Database (`credvault_cards`)

CardIssuers			
uniqueidentifier	Id	PK	
string	Name		NOT NULL (100)
int	Network		CardNetwork enum
datetime	CreatedAtUtc		
datetime	UpdatedAtUtc		

1 issuer → N cards (IssuerId FK)

CreditCards			
uniqueidentifier	Id	PK	
uniqueidentifier	UserId		Cross-service ref → identity_users.Id
uniqueidentifier	IssuerId	FK	→ CardIssuers.Id
string	CardholderName		(256)
string	Last4		(4)
string	MaskedNumber		(32)
string	EncryptedCardNumber		
int	ExpMonth		1–12
int	ExpYear		
decimal	CreditLimit		18,2
decimal	OutstandingBalance		18,2
int	BillingCycleStartDay		
bool	IsDefault		
bool	IsVerified		
datetime	VerifiedAtUtc		
bool	IsDeleted		Soft Delete flag
datetime	DeletedAtUtc		
datetime	CreatedAtUtc		
datetime	UpdatedAtUtc		

1 card → N transactions
(CardId FK)

CardTransactions			
uniqueidentifier	Id	PK	
uniqueidentifier	CardId	FK	→ CreditCards.Id
uniqueidentifier	UserId		Cross-service ref

uniqueidentifier	UserId		Cross-service ref
int	Type		1=Purchase 2=Payment 3=Refund
decimal	Amount		18,2
string	Description		(256)
datetime	DateUtc		
datetime	CreatedAtUtc		

Indexes: `CreditCards(UserId)` , `CreditCards(UserId, IsDefault)` , `CardTransactions(CardId)` , `CardTransactions(UserId)` , `CardTransactions(DateUtc)`

On Delete: CardIssuers → CreditCards: `RESTRICT` | CreditCards → CardTransactions: `RESTRICT`

7.4 Billing Database (`credvault_billing`)

RewardTiers			
uniqueidentifier	Id	PK	
int	CardNetwork		
uniqueidentifier	IssuerId		NULLABLE - cross-service
decimal	MinSpend		18,2
decimal	RewardRate		9,6 e.g. 0.015 = 1.5%
datetime	EffectiveFromUtc		
datetime	EffectiveToUtc		NULLABLE
datetime	CreatedAtUtc		
datetime	UpdatedAtUtc		

1 tier → N accounts
(RewardTierId FK RESTRICT)

RewardAccounts			
uniqueidentifier	Id	PK	
uniqueidentifier	UserId		UNIQUE - one per user
uniqueidentifier	RewardTierId	FK	→ RewardTiers.Id (NULLABLE)
decimal	PointsBalance		18,2
datetime	CreatedAtUtc		
datetime	UpdatedAtUtc		

1 account → N transactions
(RewardAccountId FK CASCADE)

Bills			
uniqueidentifier	Id	PK	
uniqueidentifier	UserId		Cross-service
uniqueidentifier	CardId		Cross-service
int	CardNetwork		
uniqueidentifier	IssuerId		Cross-service
decimal	Amount		18,2
decimal	MinDue		18,2
string	Currency		Default INR
datetime	BillingDateUtc		
datetime	DueDateUtc		
decimal	AmountPaid		NULLABLE
datetime	PaidAtUtc		NULLABLE
int	Status		1=Pending 2=Paid 3=Overdue 4=Cancelled 5=PartiallyPaid
datetime	CreatedAtUtc		
datetime	UpdatedAtUtc		

1 bill → N reward txns (BillId
FK RESTRICT nullable)

1 bill → N statements (BillId
FK RESTRICT nullable)

Statements			
uniqueidentifier	Id	PK	
uniqueidentifier	UserId		Cross-service
uniqueidentifier	CardId		Cross-service
uniqueidentifier	BillId	FK	→ Bills.Id (NULLABLE)
string	StatementPeriod		e.g. January 2024
datetime	PeriodStartUtc		
datetime	PeriodEndUtc		
datetime	GeneratedAtUtc		
datetime	DueDateUtc		
decimal	OpeningBalance		18,2
decimal	TotalPurchases		18,2
decimal	TotalPayments		18,2
decimal	TotalRefunds		18,2
decimal	PenaltyCharges		18,2
decimal	InterestCharges		18,2
decimal	ClosingBalance		18,2
decimal	MinimumDue		18,2
decimal	AmountPaid		18,2
datetime	PaidAtUtc		
int	Status		1=Generated 2=Paid 3=Overdue 4=PartiallyPaid

RewardTransactions			
uniqueidentifier	Id	PK	
uniqueidentifier	RewardAccountId	FK	→ RewardAccounts.Id
uniqueidentifier	BillId	FK	→ Bills.Id (NULLABLE)
decimal	Points		18,2
int	Type		1=Earned 2=Adjusted 3=Redeemed 4=Reversed
datetime	CreatedAtUtc		
datetime	ReversedAtUtc		NULLABLE

string	CardLast4		(10)
string	CardNetwork		(50)
string	IssuerName		(100)
decimal	CreditLimit		18,2
decimal	AvailableCredit		18,2
string	Notes		
datetime	CreatedAtUtc		
datetime	UpdatedAtUtc		

1 statement → N txns
(StatementId FK CASCADE)

StatementTransactions			
uniqueidentifier	Id	PK	
uniqueidentifier	StatementId	FK	→ Statements.Id
uniqueidentifier	SourceTransactionId		NULLABLE – cross-service ref to CardTransactions.Id
string	Type		(50)
decimal	Amount		18,2
string	Description		(256)
datetime	DateUtc		
datetime	CreatedAtUtc		

Note on `AddMissingForeignKeys` migration (2026-04-12): Two FKs were missing from the original schema and were added:

- `RewardAccounts.RewardTierId → RewardTiers.Id`
- `Statements.BillId → Bills.Id`

7.5 Payment Database (`credvault_payments`)

Payments			
uniqueidentifier	Id	PK	
uniqueidentifier	UserId		Cross-service
uniqueidentifier	CardId		Cross-service
uniqueidentifier	BillId		Cross-service
decimal	Amount		18,2
int	PaymentType		1=Full 2=Partial 3=Scheduled
int	Status		1=Initiated 2=Processing 3=Completed 4=Failed 5=Reversed 6=Cancelled
string	FailureReason		(500)
string	OtpCode		
datetime	OtpExpiresAtUtc		
datetime	CreatedAtUtc		
datetime	UpdatedAtUtc		

1 payment → N ledger entries
(PaymentId FK CASCADE)

Transactions			
uniqueidentifier	Id	PK	
uniqueidentifier	PaymentId	FK	→ Payments.Id
uniqueidentifier	UserId		
decimal	Amount		18,2
int	Type		1=Payment 2=Reversal
string	Description		(250)
datetime	CreatedAtUtc		

PaymentOrchestrationSagas			
uniqueidentifier	CorrelationId	PK	MassTransit correlation
string	CurrentState		(64)
uniqueidentifier	PaymentId		
uniqueidentifier	UserId		
string	Email		(256)
string	FullName		(256)
uniqueidentifier	CardId		
uniqueidentifier	BillId		
decimal	Amount		18,2
string	PaymentType		(32)
decimal	RewardsAmount		18,2
bool	RewardsRedeemed		
string	OtpCode		
datetime	OtpExpiresAtUtc		
bool	OtpVerified		
bool	PaymentProcessed		
bool	BillUpdated		
bool	CardDeducted		
string	PaymentError		(500)
string	BillUpdateError		(500)
string	CardDeductionError		(500)
string	CompensationReason		(500)
int	CompensationAttempts		
datetime	CreatedAtUtc		
datetime	UpdatedAtUtc		

Indexes: `Payments(UserId)` , `Payments(BillId)` , `Transactions(UserId)`

7.6 Notification Database (`credvault_notifications`)

AuditLogs			
uniqueidentifier	Id	PK	
string	EntityName		(100)
string	EntityId		
string	Action		Created|Updated|Deleted
string	UserId		(100) NULLABLE
string	Changes		JSON
string	TraceId		(128)
datetime	CreatedAtUtc		

NotificationLogs			
uniqueidentifier	Id	PK	
uniqueidentifier	UserId		NULLABLE
string	Recipient		(255)
string	Subject		(500)
string	Body		
string	Type		email|sms
bool	IsSuccess		
string	ErrorMessage		
string	TraceId		(128)
datetime	CreatedAtUtc		

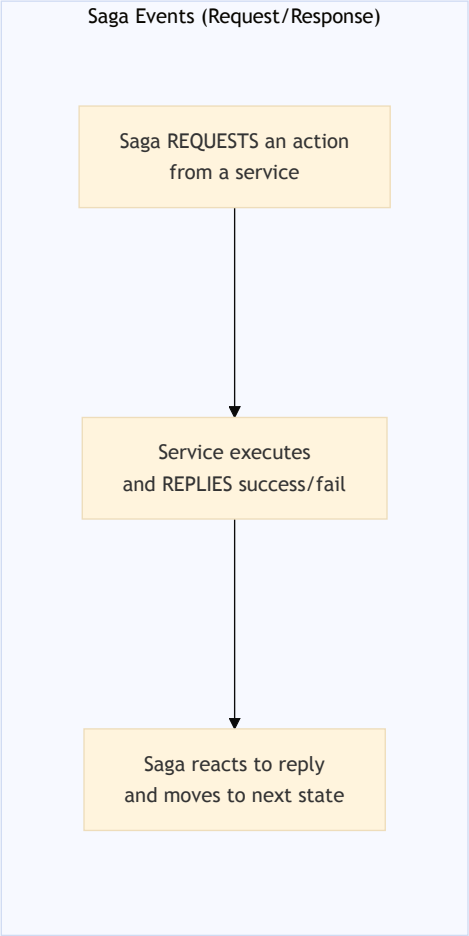
7.7 Cross-Service Reference Map

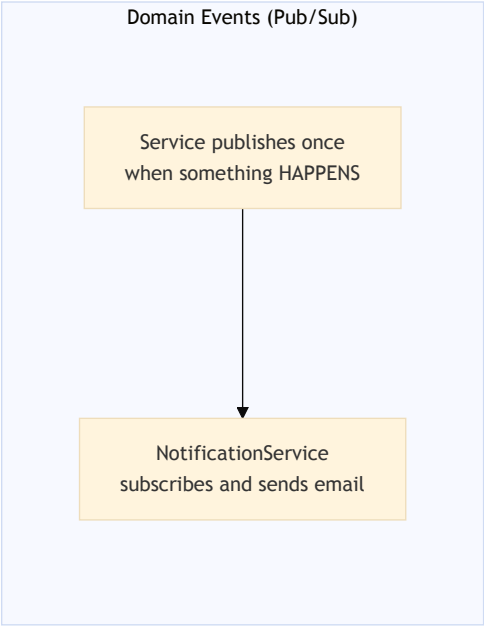
SQL foreign key constraints **cannot** cross databases. The table below documents all logical cross-service references and how they are validated at the application layer.

Column	Host Table	Host DB	Logically References	Owner DB	Validation Method
`UserId`	`CreditCards`	cards	`identity_users.Id`	identity	JWT claim check
`UserId`	`Bills`	billing	`identity_users.Id`	identity	JWT claim + event payload
`CardId`	`Bills`	billing	`CreditCards.Id`	cards	Event payload from card service
`IssuerId`	`Bills`	billing	`CardIssuers.Id`	cards	Stored at bill generation time
`CardId`	`Statements`	billing	`CreditCards.Id`	cards	Bill generation context
`UserId`	`Statements`	billing	`identity_users.Id`	identity	JWT claim
`IssuerId`	`RewardTiers`	billing	`CardIssuers.Id`	cards	Admin-managed data entry
`SourceTransactionId`	`StatementTransactions`	billing	`CardTransactions.Id`	cards	Best-effort denormalization
`CardId`	`Payments`	payments	`CreditCards.Id`	cards	API call at payment initiation
`BillId`	`Payments`	payments	`Bills.Id`	billing	API call at payment initiation
`UserId`	`Payments`	payments	`identity_users.Id`	identity	JWT claim
`UserId`	`NotificationLogs`	notifications	`identity_users.Id`	identity	Event payload

8. Event-Driven Architecture (RabbitMQ)

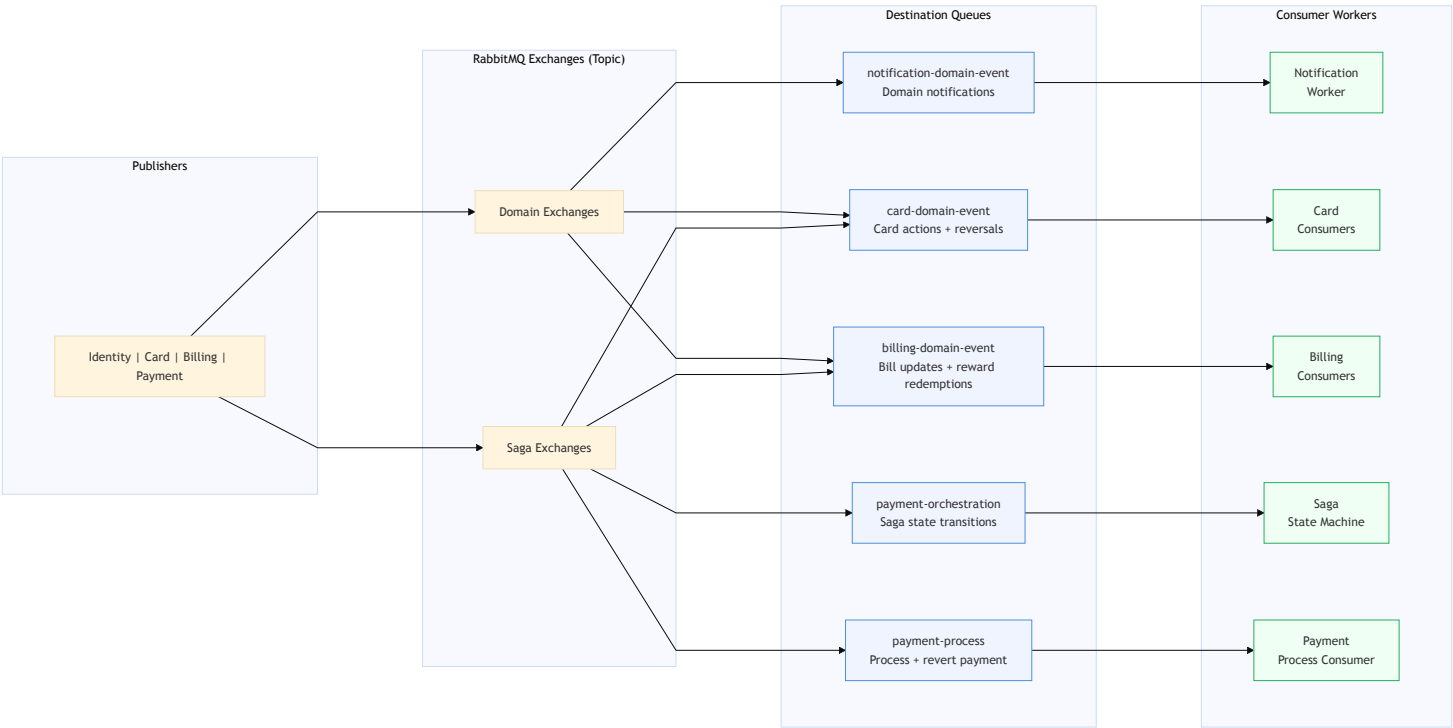
8.1 Two Event Patterns in CredVault





Attribute	Domain Events	Saga Events
Pattern	Fire and forget (Pub/Sub)	Request → Response
Consumer	Notification Service (emails)	Always the Saga
Routing	Fanout via exchange	Direct point-to-point queue
Failure Handling	DLQ after retries	Saga compensation logic
Count	10 events	22 events

8.2 Queue Architecture



8.3 Retry & Dead Letter Policy

Stage	Behavior
Attempt 1	Message processed immediately
Attempt 2	Retry after 5 seconds (fast retry)
Attempt 3	Retry after 30 seconds (slow retry)
Exhausted	Message moved to `<queue-name>.error` DLQ
Saga Fault	Max 3 internal attempts → transitions to `Compensating` state

9. Complete Event Catalog

9.1 Summary

Category	Count	Working
Identity Events	3	2 active, 1 scheduled for activation

Category	Count	Working
Card Events	1	✅
Billing Events	2	1 active, 1 pending downstream adoption
Payment Domain Events	4	✅
Saga Orchestration Events	22	✅
Total	**32**	**28 active in current release**

9.2 Identity Events

File: `shared.contracts/Shared.Contracts/Events/Identity/IdentityEvents.cs`

Event	Publisher	Consumer(s)	Properties
`IUserRegistered`	RegisterCommandHandler	NotificationService	UserId, Email, FullName, CreatedAtUtc
`IUserOtpGenerated`	RegisterHandler, ResendVerificationHandler, ForgotPasswordHandler	NotificationService	UserId, Email, FullName, OtpCode, Purpose, ExpiresAtUtc
`IUserDeleted`	Identity user lifecycle workflow (activation pending)	Card, Billing, Payment	UserId, DeletedAtUtc

9.3 Card Events

File: `shared.contracts/Shared.Contracts/Events/Card/CardEvents.cs`

Event	Publisher	Consumer(s)	Properties
`ICardAdded`	CreateCardCommandHandler	NotificationService	CardId, UserId, Email, FullName, CardNumberLast4, CardHolderName, AddedAt

9.4 Billing Events

File: `shared.contracts/Shared.Contracts/Events/Billing/BillingEvents.cs`

Event	Publisher	Consumer(s)	Properties
`IBillGenerated`	GenerateBillCommandHandler	NotificationService	BillId, UserId, Email, FullName, CardId, Amount, DueDate, GeneratedAt
`IBillOverdueDetected`	CheckOverdueBillsCommandHandler	Planned: Notification reminders, Billing penalty processor	BillId, CardId, UserId, OverdueAmount, DueDate, DaysOverdue, DetectedAt

9.5 Payment Domain Events

File: `shared.contracts/Shared.Contracts/Events/Payment/PaymentEvents.cs`

Event	Publisher	Consumer(s)	Properties
`IPaymentCompleted`	PaymentOrchestrationSaga (on completion)	NotificationService, Saga	PaymentId, UserId, Email, FullName, CardId, BillId, Amount, AmountPaid, RewardsRedeemed, CompletedAt
`IPaymentFailed`	PaymentOrchestrationSaga (on compensation)	NotificationService, Saga	PaymentId, UserId, Email, FullName, Amount, Reason, FailedAt
`IPaymentReversed`	PaymentOrchestrationSaga (reversal)	Card Service (restores balance)	PaymentId, UserId, BillId, CardId, Amount, PointsDeducted, ReversedAt

Event	Publisher	Consumer(s)	Properties
`IPaymentOtpGenerated`	InitiatePaymentCommandHandler	NotificationService	PaymentId, UserId, Email, FullName, Amount, OtpCode, ExpiresAtUtc

9.6 Saga Orchestration Events

File: `shared.contracts/Shared.Contracts/Events/Saga/SagaOrchestrationEvents.cs`

Saga → Services (Commands):

Event	Trigger	Target
`IStartPaymentOrchestration`	Payment initiated	Saga starts
`IOtpVerified`	User submits correct OTP	Saga transitions
`IOtpFailed`	User submits wrong OTP	Saga marks failure
`IPaymentProcessRequested`	OTP verified	PaymentProcessConsumer
`IBillUpdateRequested`	Payment processed	Billing SagaConsumer
`ICardDeductionRequested`	Bill updated	Card SagaConsumer
`IRewardRedemptionRequested`	Bill updated (if rewards)	RewardRedemptionConsumer
`IRevertBillUpdateRequested`	Card deduction failed	Billing SagaConsumer (compensate)
`IRevertPaymentRequested`	Bill revert succeeded	RevertPaymentConsumer

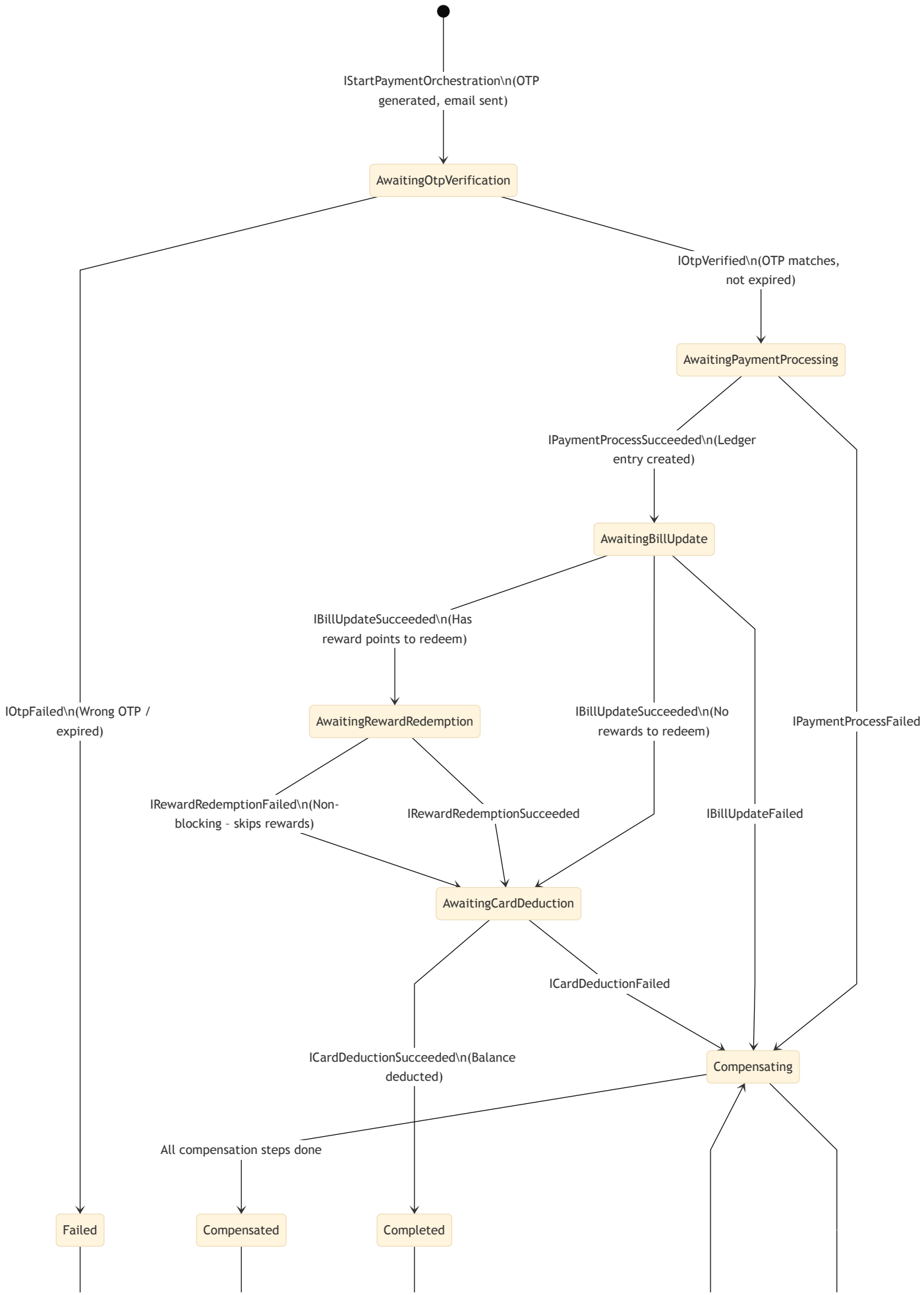
Services → Saga (Responses):

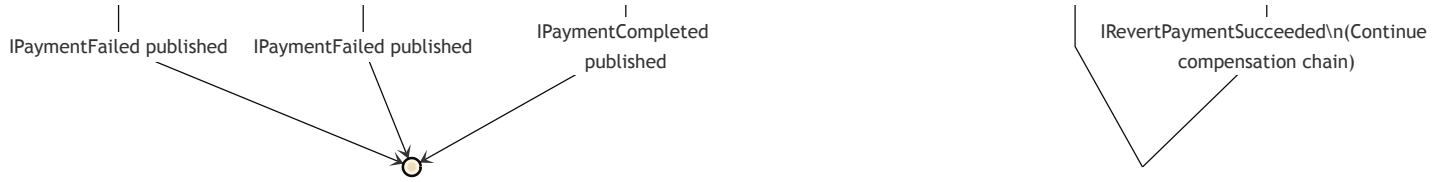
Event	Sender	Meaning
`IPaymentProcessSucceeded`	PaymentProcessConsumer	Payment ledger created
`IPaymentProcessFailed`	PaymentProcessConsumer	Ledger failed
`IBillUpdateSucceeded`	BillUpdateSagaConsumer	Bill marked processing
`IBillUpdateFailed`	BillUpdateSagaConsumer	Bill update error
`ICardDeductionSucceeded`	CardDeductionSagaConsumer	Balance deducted
`ICardDeductionFailed`	CardDeductionSagaConsumer	Insufficient balance / error
`IRevertBillUpdateSucceeded`	RevertBillUpdateConsumer	Bill reverted to Pending
`IRevertBillUpdateFailed`	RevertBillUpdateConsumer	Revert failed
`IRevertPaymentSucceeded`	RevertPaymentConsumer	Payment voided
`IRevertPaymentFailed`	RevertPaymentConsumer	Void failed
`IRewardRedemptionSucceeded`	RewardRedemptionConsumer	Points deducted
`IRewardRedemptionFailed`	RewardRedemptionConsumer	Points deduction error

10. Saga State Machine — Payment Orchestration

The Payment Orchestration Saga coordinates a distributed, multi-service atomic operation. It is implemented as a MassTransit `MassTransitStateMachine<PaymentOrchestrationSagaState>` .

10.1 State Transition Diagram


IStartPaymentOrchestration\n(OTP generated, email sent)
AwaitingOtpVerification
IOtpVerified\n(OTP matches, not expired)
IOtpFailed\n(Wrong OTP / expired)
AwaitingPaymentProcessing
IPaymentProcessSucceeded\n(Ledger entry created)
IPaymentProcessFailed
AwaitingBillUpdate
IBillUpdateSucceeded\n(Has reward points to redeem)
IBillUpdateSucceeded\n(No rewards to redeem)
IBillUpdateFailed
AwaitingRewardRedemption
IRewardRedemptionFailed\n(Non-blocking - skips rewards)
IRewardRedemptionSucceeded
AwaitingCardDeduction
ICardDeductionFailed
ICardDeductionSucceeded\n(Balance deducted)
Compensating
All compensation steps done
Failed
Compensated
Completed



10.2 State Descriptions

State	Description	Next Trigger
`AwaitingOtpVerification`	Saga created. OTP emailed to user. Waiting for `/verify-otp` call.	`IOtpVerified` or `IOtpFailed`
`AwaitingPaymentProcessing`	OTP validated. Payment process consumer asked to create ledger entry.	`IPaymentProcessSucceeded` or `IPaymentProcessFailed`
`AwaitingBillUpdate`	Ledger entry created. Billing asked to mark bill as paying.	`IBillUpdateSucceeded` or `IBillUpdateFailed`
`AwaitingRewardRedemption`	Bill updated. If user has reward points selected, redemption is requested.	`IRewardRedemptionSucceeded` or `IRewardRedemptionFailed`
`AwaitingCardDeduction`	Rewards handled. Card service asked to deduct outstanding balance.	`ICardDeductionSucceeded` or `ICardDeductionFailed`
`Completed`	All steps succeeded. `IPaymentCompleted` published.	Terminal
`Compensating`	A step failed. Rolling back completed steps in reverse order.	Compensation responses
`Compensated`	All rollback steps done. `IPaymentFailed` published.	Terminal
`Failed`	OTP failure path. No compensation needed.	Terminal

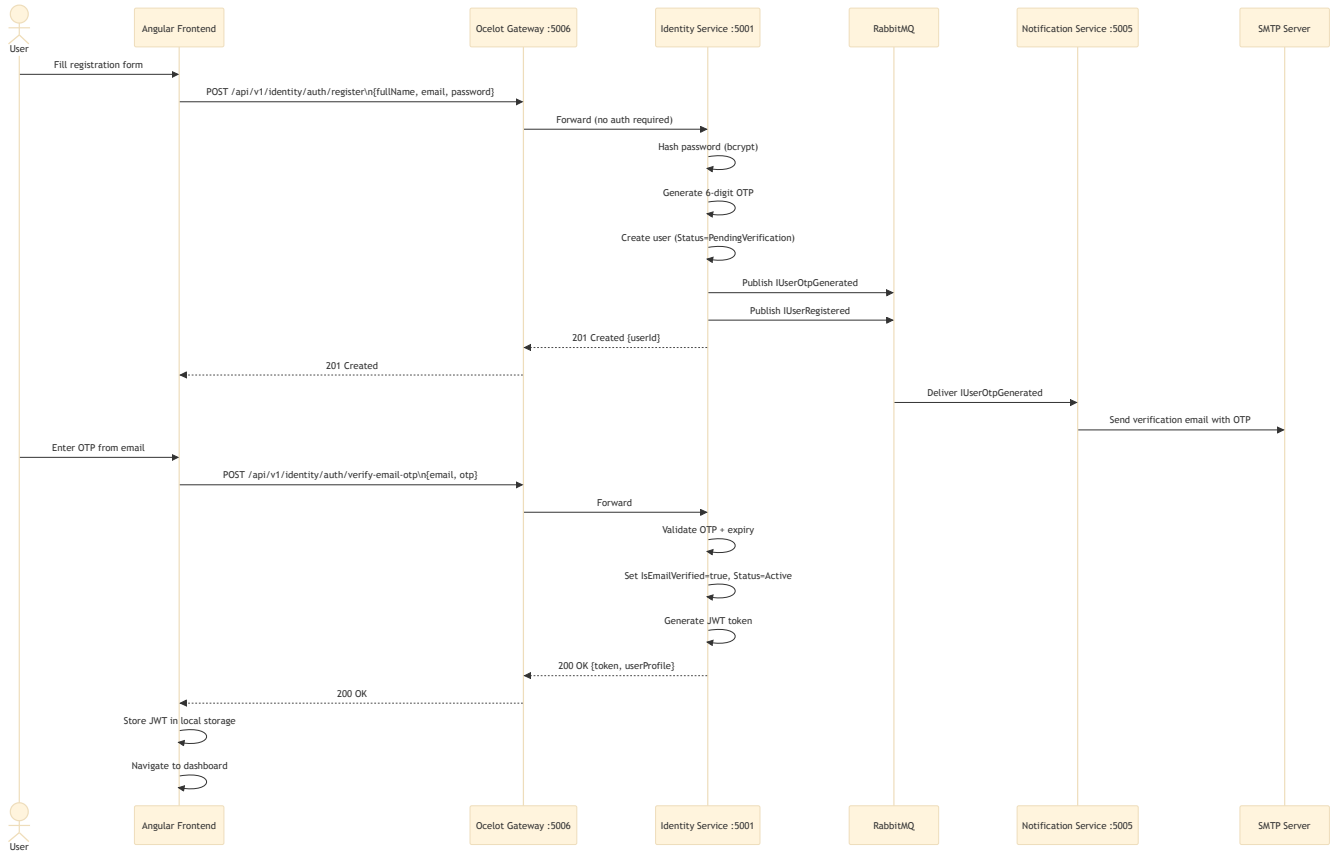
10.3 Compensation Chain

Compensation executes in reverse order of execution:

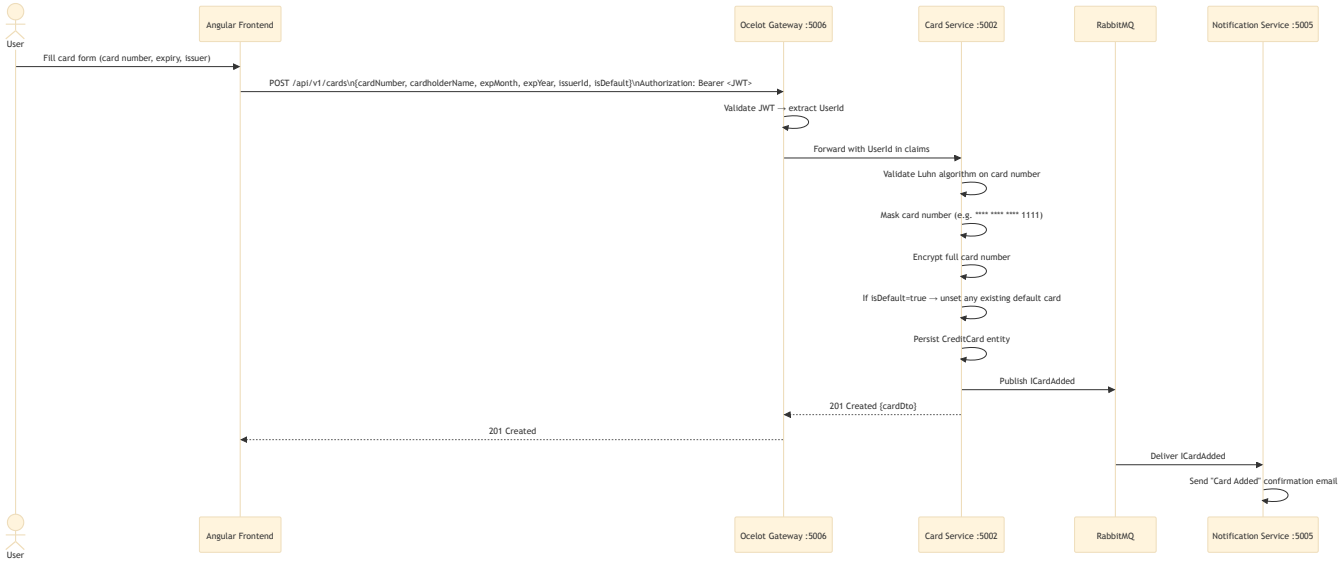
```
Step that failed: ICardDeductionFailed
    ↓
Compensate Step 3: Publish IRevertBillUpdateRequested → Billing reverts bill to Pending
    ↓
Compensate Step 2: Publish IRevertPaymentRequested → Payment voids ledger entry
    ↓
State → Compensated
    ↓
Publish IPaymentFailed
```

11. Sequence Diagrams — Core User Flows

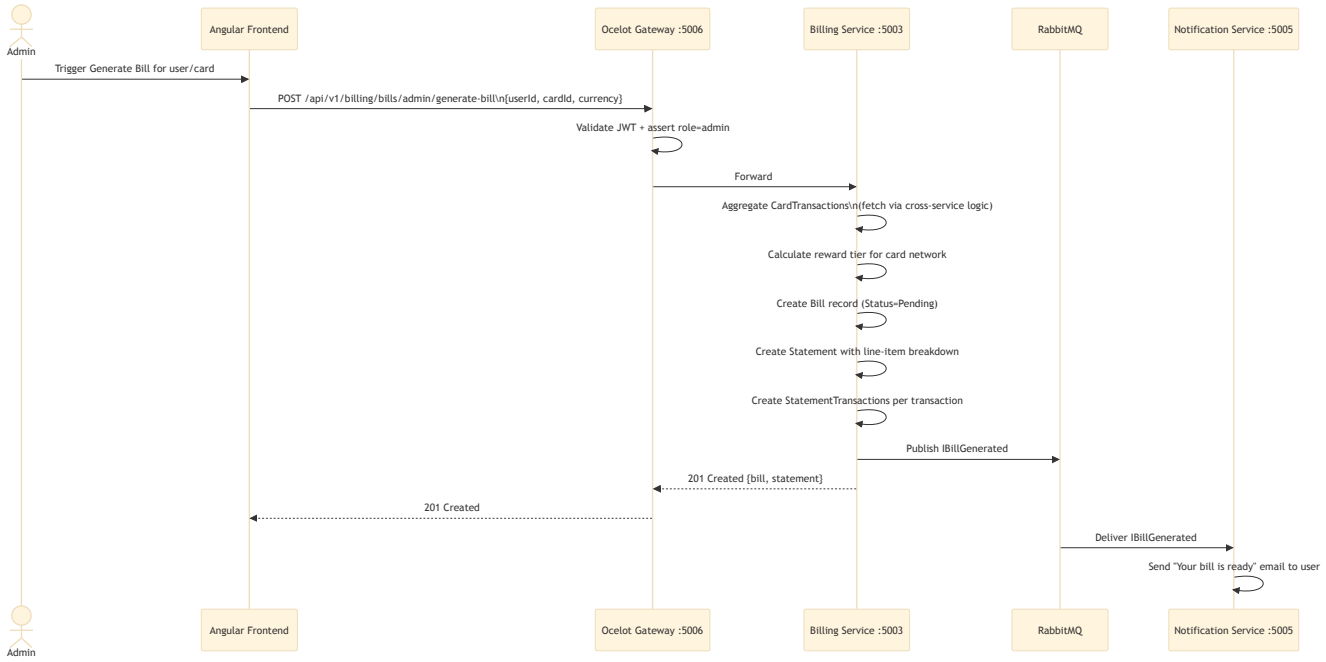
11.1 User Registration & Email Verification



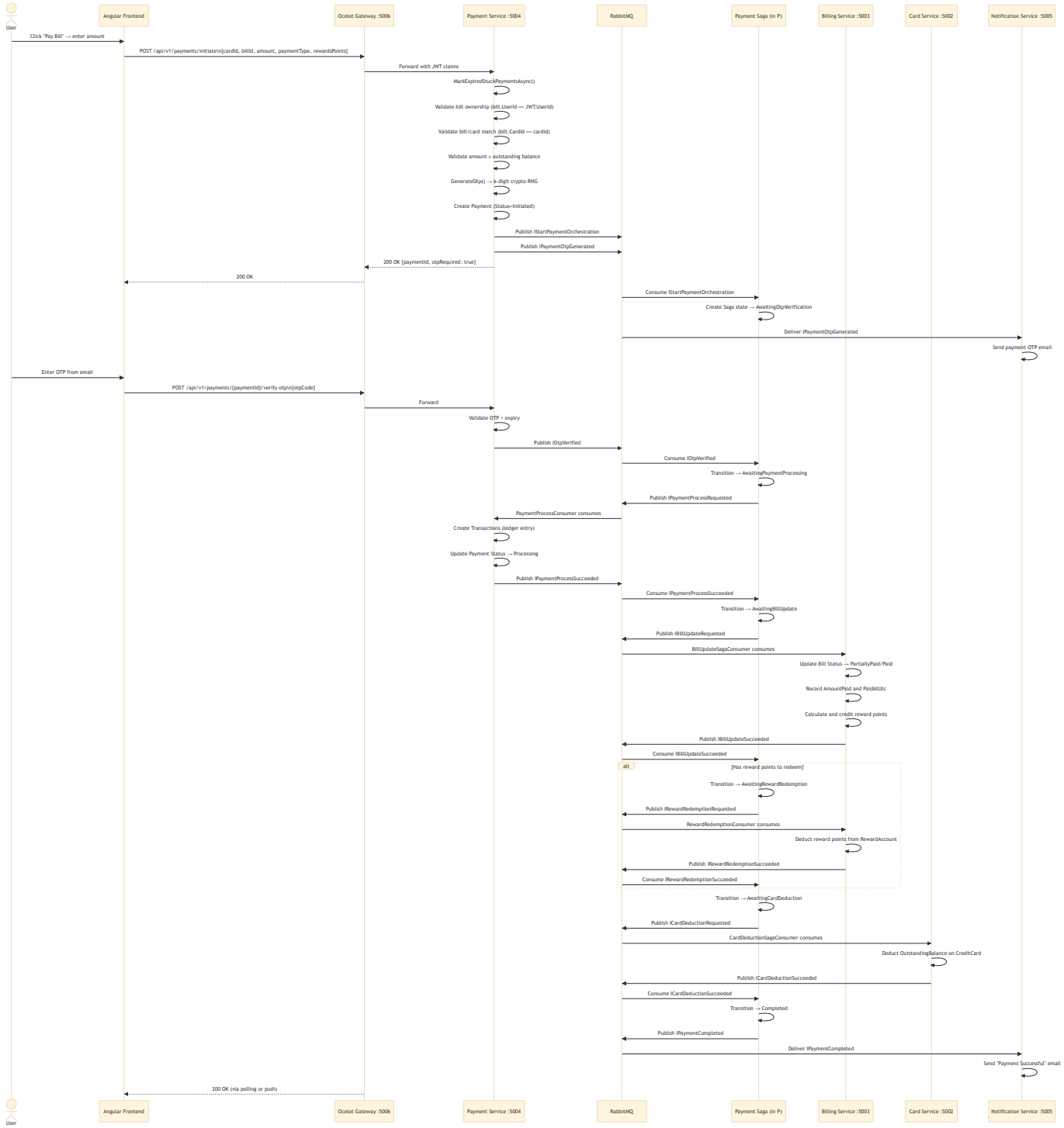
11.2 Add Credit Card



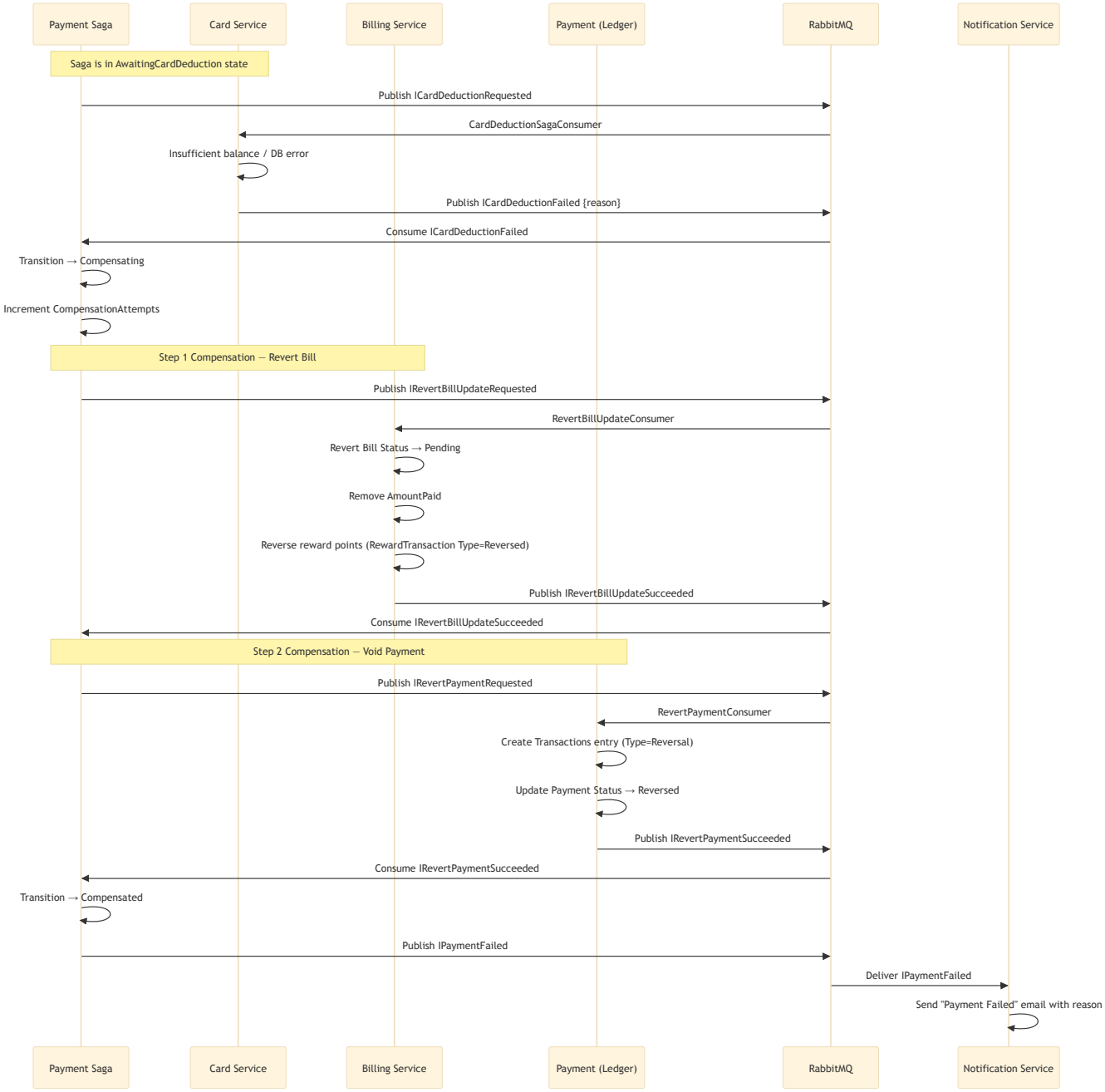
11.3 Bill Generation (Admin)



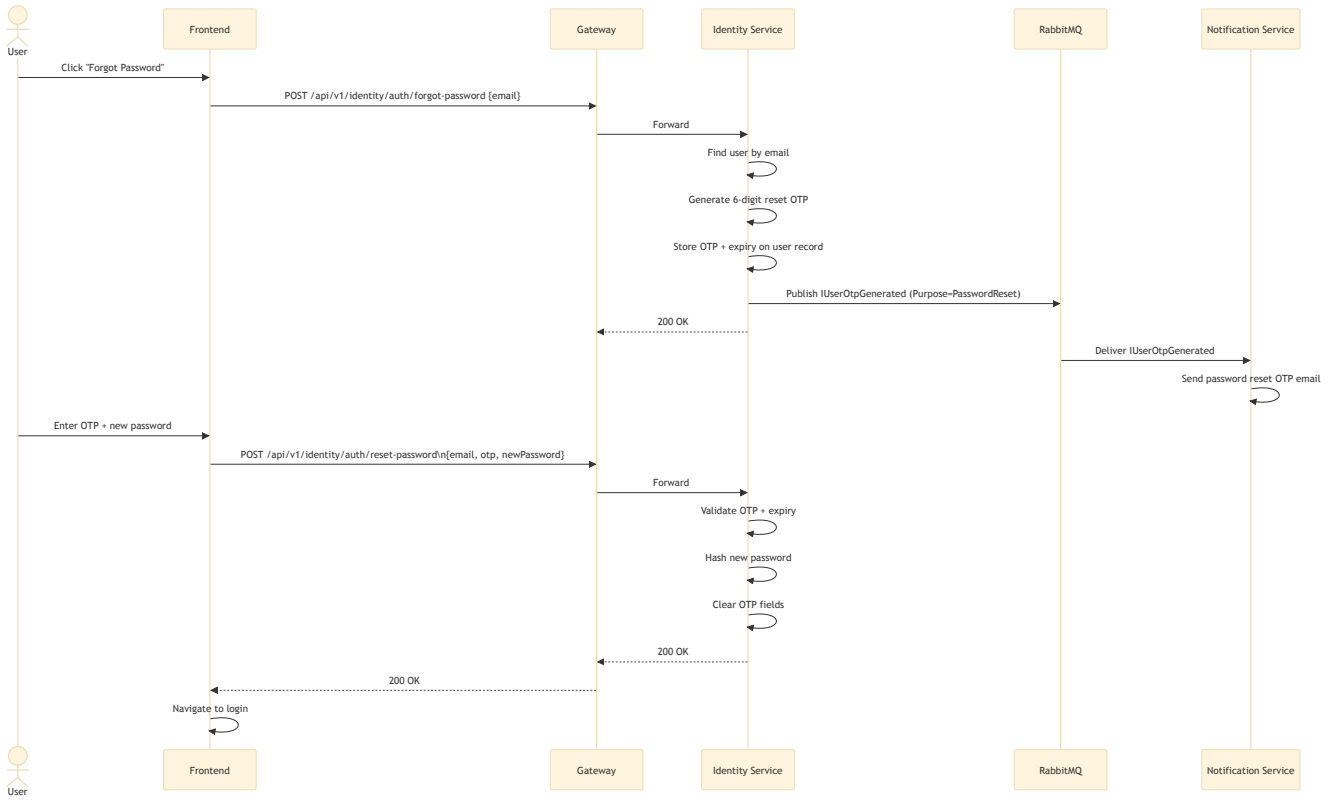
11.4 Payment Flow — Happy Path (Full Saga)



11.5 Payment Flow — Failure & Compensation



11.6 Password Reset Flow



12. API Specification

All external traffic routes through the Ocelot API Gateway at `:5006` . The gateway enforces JWT Bearer validation and forwards requests to the appropriate upstream service.

12.1 Authentication Matrix

Access Level	Token Required	Role Claim
Public	No	—
Authenticated	Yes (`Bearer <JWT>`)	Any valid role
Admin	Yes (`Bearer <JWT>`)	`role: admin`

12.2 Identity Service — `/api/v1/identity`

Auth Endpoints

Method	Path	Auth	Description
POST	`/auth/register`	Public	Register new user
POST	`/auth/login`	Public	Login with email/password
POST	`/auth/google`	Public	Google OAuth login
POST	`/auth/resend-verification`	Public	Resend email OTP
POST	`/auth/verify-email-otp`	Public	Verify email with OTP

Method	Path	Auth	Description
POST	`/auth/forgot-password`	Public	Trigger password reset OTP
POST	`/auth/reset-password`	Public	Reset password with OTP

User Endpoints

Method	Path	Auth	Description
GET	`/users/me`	Authenticated	Get own profile
PUT	`/users/me`	Authenticated	Update own profile
PUT	`/users/me/password`	Authenticated	Change own password
GET	`/users`	Admin	Paginated user list
GET	`/users/stats`	Admin	Aggregate user stats
GET	`/users/{userId}`	Admin	Get user by ID
PUT	`/users/{userId}/status`	Admin	Update user status
PUT	`/users/{userId}/role`	Admin	Update user role

12.3 Card Service — `/api/v1/cards`

Method	Path	Auth	Description
POST	`/cards`	Authenticated	Add a new credit card
GET	`/cards`	Authenticated	List my cards
GET	`/cards/transactions`	Authenticated	All my transactions (all cards)
GET	`/cards/{cardId}`	Authenticated	Card details
PUT	`/cards/{cardId}`	Authenticated	Update card details
DELETE	`/cards/{cardId}`	Authenticated	Soft-delete card
GET	`/cards/{cardId}/transactions`	Authenticated	Transactions for a card
POST	`/cards/{cardId}/transactions`	Authenticated	Add a transaction
GET	`/cards/admin/{cardId}`	Admin	Admin: card details
GET	`/cards/admin/{cardId}/transactions`	Admin	Admin: card transactions
PUT	`/cards/{cardId}/admin`	Admin	Admin: update card (limits, balance)
GET	`/cards/user/{userId}`	Admin	Admin: get cards by user
GET	`/issuers`	Authenticated	List card issuers
POST	`/issuers`	Admin	Create card issuer
PUT	`/issuers/{id}`	Admin	Update card issuer
DELETE	`/issuers/{id}`	Admin	Delete card issuer

12.4 Billing Service — `/api/v1/billing`

Method	Path	Auth	Description
GET	`/bills`	Authenticated	List my bills
GET	`/bills/{id}`	Authenticated	Bill details

Method	Path	Auth	Description
POST	`/bills/admin/generate-bill`	Admin	Generate a bill for user/card
POST	`/bills/admin/check-overdue`	Admin	Trigger overdue detection
GET	`/bills?userId=`	Admin	Get all bills for a user
GET	`/bills/has-pending/{cardId}`	Public	Check pending bill exists
GET	`/statements`	Authenticated	List my statements
GET	`/statements/{id}`	Authenticated	Statement details
GET	`/statements?userId=&cardId=`	Admin	Admin: statements filter
GET	`/statements/admin/all`	Admin	All statements
GET	`/statements/admin/{statementId}/full`	Admin	Full statement detail
GET	`/rewards/tiers`	Authenticated	List reward tiers
POST	`/rewards/tiers`	Admin	Create reward tier
PUT	`/rewards/tiers/{id}`	Admin	Update reward tier
DELETE	`/rewards/tiers/{id}`	Admin	Delete reward tier
GET	`/rewards/account`	Authenticated	My reward account & balance
POST	`/rewards/redeem`	Authenticated	Redeem reward points
GET	`/rewards/transactions?userId=`	Admin	Reward transaction history
POST	`/rewards/internal/redeem`	Public (internal)	Internal reward deduction endpoint

12.5 Payment Service — `/api/v1/payments`

Method	Path	Auth	Description
POST	`/payments/initiate`	Authenticated	Initiate payment (spawns saga)
POST	`/payments/{paymentId}/verify-otp`	Authenticated	Submit OTP to trigger saga
POST	`/payments/{paymentId}/resend-otp`	Authenticated	Resend payment OTP
GET	`/payments`	Authenticated	My payment history
GET	`/payments/{id}`	Authenticated	Payment details

12.6 Notification Service — `/api/v1/notifications`

Method	Path	Auth	Description
GET	`/notifications/logs`	Admin	View notification send logs
GET	`/notifications/audit`	Admin	View audit log entries

13. Request & Response Contracts

13.1 Standard Response Envelope

Every API response is wrapped in `ApiResponse<T>` :

```
{
  "success": true,
  "message": "Card added successfully",
  "errorCode": null,
  "data": {
    "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
  },
  "traceId": "0HN5PBZXK3AGT:00000001"
}
```

Error Response:

```
{
  "success": false,
  "message": "Bill amount has already been fully paid",
  "errorCode": "BILL_ALREADY_PAID",
  "data": null,
  "traceId": "0HN5PBZXK3AGT:00000002"
}
```

13.2 Key Request Contracts

Register User:

```
{
  "fullName": "Aarav Sharma",
  "email": "aarav@example.com",
  "password": "StrongPassword@123"
}
```

Login:

```
{
  "email": "aarav@example.com",
  "password": "StrongPassword@123"
}
```

Add Card:

```
{
  "cardholderName": "AARAV SHARMA",
  "cardNumber": "4111111111111111",
  "expMonth": 12,
  "expYear": 2028,
  "issuerId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "isDefault": true
}
```

Add Transaction:

```
{
  "type": 1,
  "amount": 1250.00,
  "description": "Restaurant - Zomato",
  "dateUtc": "2026-04-10T18:30:00Z"
}
```

Initiate Payment:

```
{
  "cardId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "billId": "7cb85f64-5717-4562-b3fc-2c963f66afb7",
  "amount": 2500.00,
  "paymentType": "Full",
  "rewardsPoints": 50
}
```

Verify Payment OTP:

```
{
  "otpCode": "483920"
}
```

Redeem Rewards:

```
{
  "points": 100,
  "target": "Bill",
  "billId": "7cb85f64-5717-4562-b3fc-2c963f66afb7"
}
```

13.3 Key Response Contracts

Login Response:

```
{
  "success": true,
  "message": "Login successful",
  "data": {
    "token": "eyJhbGciOiJIUzI1NiIs...",
    "userId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "email": "aarav@example.com",
    "fullName": "Aarav Sharma",
    "role": "user"
  }
}
```

Card Response:

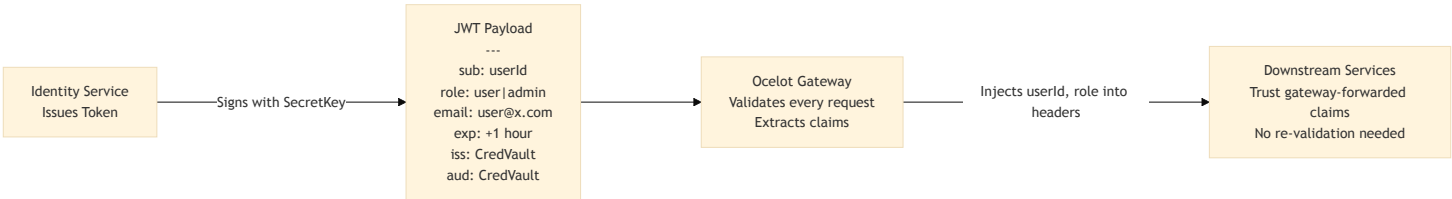
```
{
  "success": true,
  "data": {
    "id": "...",
    "cardholderName": "AARAV SHARMA",
    "maskedNumber": "**** **** **** 1111",
    "last4": "1111",
    "network": "Visa",
    "issuerName": "HDFC Bank",
    "creditLimit": 100000.00,
    "outstandingBalance": 5000.00,
    "isDefault": true,
    "isVerified": false
  }
}
```

Bill Response:

```
{
  "success": true,
  "data": {
    "id": "...",
    "amount": 5000.00,
    "minDue": 500.00,
    "currency": "INR",
    "billingDate": "2026-04-01",
    "dueDate": "2026-04-20",
    "status": "Pending",
    "amountPaid": null
  }
}
```

14. Security Architecture

14.1 JWT Token Structure



14.2 Security Controls per Endpoint Type

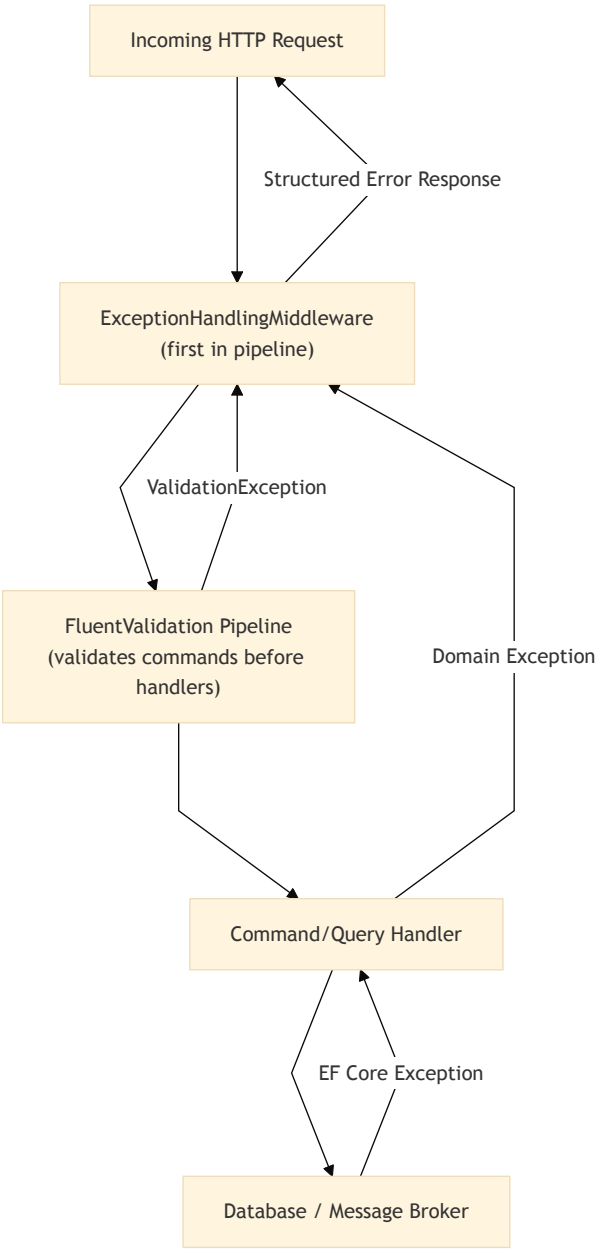
Concern	Implementation
Authentication	JWT Bearer token validated at Ocelot gateway
Authorization	`[Authorize(Roles = "admin")]` on admin controllers
Password Storage	BCrypt hashed, never stored in plaintext
Payment Authorization	6-digit OTP (crypto-RNG), 5-minute TTL, single-use
Card Data	Full card number AES-encrypted at rest; only `Last4` + `MaskedNumber` stored in plaintext
OTP Expiry	`OtpExpiresAtUtc` checked server-side; expired OTPs rejected
Stack Trace Leaking	`ExceptionHandlingMiddleware` catches all unhandled exceptions, logs internally, returns generic `500` to client
Input Validation	FluentValidation pipeline validates all command inputs before handlers execute
Stuck Payment Cleanup	`MarkExpiredStuckPaymentsAsync()` runs on every new payment initiation to clear zombie saga states

14.3 OTP Flow Detail

```
1. Crypto-RNG generates 6-digit code
2. Code stored on entity (User or Payment record) with 5-minute expiry
3. Code published in event → Notification Service → Email
4. On verification:
   a. Code retrieved from DB
   b. Current UTC compared against expiry
   c. Code compared (constant-time string comparison)
   d. On success: code cleared from DB (single-use enforcement)
```

15. Error Handling & Exception Strategy

15.1 Exception Middleware Pipeline



15.2 HTTP Status Code Map

HTTP Status	Scenario	Example
`200 OK`	Successful read or action	Login, get cards
`201 Created`	New resource created	Register, add card
`400 Bad Request`	Validation failure, business rule violation	Amount > outstanding balance
`401 Unauthorized`	No token / invalid / expired JWT	Missing `Authorization` header
`403 Forbidden`	Valid token but insufficient role	User accessing admin endpoint
`404 Not Found`	Entity not found in DB	`GET /cards/{id}` for unknown ID
`409 Conflict`	Duplicate resource	Registering with existing email

HTTP Status	Scenario	Example
`503 Service Unavailable`	Downstream dependency unavailable	RabbitMQ down
`500 Internal Server Error`	Unhandled exception (middleware catches)	DB connection failure

15.3 Domain Error Codes

Error Code	Meaning
`INVALID_OTP`	OTP does not match
`EXPIRED_OTP`	OTP TTL exceeded
`BILL_NOT_FOUND`	Bill ID not found
`BILL_ALREADY_PAID`	Bill status is Paid
`BILL_CARD_MISMATCH`	Bill.CardId ≠ requested CardId
`OVERPAYMENT`	Amount > outstanding balance
`CARD_NOT_FOUND`	Card ID not found
`INVALID_LUHN`	Card number fails Luhn check
`USER_SUSPENDED`	Login blocked due to status
`EMAIL_NOT_VERIFIED`	Login blocked until email verified

16. Infrastructure & Deployment

16.1 Service Port Map

Service	Internal Port	External (Docker)
Identity Service	5001	5001
Card Service	5002	5002
Billing Service	5003	5003
Payment Service	5004	5004
Notification Service	5005	5005
API Gateway (Ocelot)	5006	5006
SQL Server	1433	1434
RabbitMQ AMQP	5672	5672
RabbitMQ Management UI	15672	15672
Angular Frontend	4200	4200

16.2 Environment Variables

```
# Database Connections
IDENTITY_DB_CONNECTION=Server=localhost,1434;Database=credvault_identity;User Id=sa;Password=...
CARD_DB_CONNECTION=Server=localhost,1434;Database=credvault_cards;User Id=sa;Password=...
BILLING_DB_CONNECTION=Server=localhost,1434;Database=credvault_billing;User Id=sa;Password=...
PAYMENT_DB_CONNECTION=Server=localhost,1434;Database=credvault_payments;User Id=sa;Password=...

# JWT
JwtOptions__SecretKey=<min-32-char-secret>
JwtOptions__Issuer=CredVault
JwtOptions__Audience=CredVault
JwtOptions__ExpiryMinutes=60

# RabbitMQ
RabbitMqSettings__Host=localhost
RabbitMqSettings__Port=5672
RabbitMqSettings__Username=guest
RabbitMqSettings__Password=guest

# SMTP
SmtpSettings__Host=smtp.gmail.com
SmtpSettings__Port=587
SmtpSettings__Username=credvault@gmail.com
SmtpSettings__Password=<app-password>
SmtpSettings__FromName=CredVault
```

16.3 Migration Locations

```
server/services/
├── identity-service/IdentityService.Infrastructure/Persistence/Sql/Migrations/
├── card-service/CardService.Infrastructure/Persistence/Sql/Migrations/
├── billing-service/BillingService.Infrastructure/Persistence/Sql/Migrations/
│   └── 20260412100953_AddMissingForeignKeys.cs  ← Latest
├── payment-service/PaymentService.Infrastructure/Persistence/Sql/Migrations/
└── notification-service/NotificationService.Infrastructure/Migrations/
```

16.4 Health Check Endpoints

```
curl http://localhost:5006/health                       # Gateway
curl http://localhost:5006/api/v1/identity/health       # Identity
curl http://localhost:5006/api/v1/cards/health          # Card
curl http://localhost:5006/api/v1/billing/health        # Billing
curl http://localhost:5006/api/v1/payments/health       # Payment
```

16.5 Diagnostic SQL Queries

```sql
-- Find stuck/zombie payments (initiated > 1 hour ago with no saga completion)
SELECT * FROM payments
WHERE Status = 1
AND CreatedAtUtc < DATEADD(hour, -1, GETUTCDATE());

-- Find saga states that never completed
SELECT CorrelationId, CurrentState, CreatedAtUtc, Amount
FROM payment_orchestration_saga_state
WHERE CurrentState NOT IN ('Completed', 'Compensated', 'Failed')
ORDER BY CreatedAtUtc DESC;

-- Check overdue bills
SELECT Id, UserId, Amount, DueDateUtc, Status
FROM bills
WHERE Status = 1  -- Pending
AND DueDateUtc < GETUTCDATE();

-- Reward points summary per user
SELECT UserId, PointsBalance, UpdatedAtUtc
FROM reward_accounts
ORDER BY PointsBalance DESC;
```

17. Known Issues & Technical Debt

17.1 High Priority: Activate `IUserDeleted` Lifecycle Event

Priority: High

Description: Admin status updates can mark a user as `Deleted`, but the Identity Service does not yet emit `IUserDeleted` in the current release path. Card, Billing, and Payment consumers are already subscribed and ready to process the event.

Impact:

Service	Consumer	Orphaned Data
Card Service	`UserDeletedConsumer`	CreditCards, CardTransactions remain
Billing Service	`UserDeletedConsumer`	Bills, Statements, RewardAccounts remain
Payment Service	*(implied)*	Payments, Sagas remain

Current Gap: A dedicated deletion command flow, delete endpoint, and event emission path are not yet part of the Identity Service API surface.

Implementation Plan:

1. Add `DeleteUserCommandHandler` in Identity → Application → Commands
2. Add `DELETE /api/v1/identity/users/{id}` (Admin only)
3. Handler sets `Status = Deleted` then publishes `IUserDeleted`

17.2 Medium Priority: Connect `IBillOverdueDetected` Consumers

Priority: Medium

Description: `CheckOverdueBillsCommandHandler` in Billing Service detects overdue bills and publishes `IBillOverdueDetected`. Downstream consumer subscriptions are pending integration.

Impact: Users with overdue bills do not receive reminder emails. No late fee application is triggered.

Potential Consumers to Add:

Consumer	Action
Notification Service	Send "Your bill is overdue" email
Billing Service (self)	Auto-apply penalty charges

17.3 Low Priority: Notification Controller — Role Enforcement

Severity: 🟡 Low

Description: The Notification controller is behind JWT authentication, but `[Authorize(Roles = "admin")]` attribute is not applied to action methods. Any authenticated user who discovers the endpoint URL can query notification and audit logs.

Fix: Add `[Authorize(Roles = "admin")]` to `NotificationLogsController` and `AuditLogsController`.

Appendix A — Shared Enum Reference

Enum	Values
`CardNetwork`	Unknown(0), Visa(1), Mastercard(2)
`UserStatus`	PendingVerification(1), Active(2), Suspended(3), Blocked(4), Deleted(5)
`UserRole`	User(1), Admin(2)
`TransactionType`	Purchase(1), Payment(2), Refund(3)
`BillStatus`	Pending(1), Paid(2), Overdue(3), Cancelled(4), PartiallyPaid(5)
`PaymentStatus`	Initiated(1), Processing(2), Completed(3), Failed(4), Reversed(5), Cancelled(6)
`PaymentType`	Full(1), Partial(2), Scheduled(3)
`RewardTransactionType`	Earned(1), Adjusted(2), Redeemed(3), Reversed(4)
`StatementStatus`	Generated(1), Paid(2), Overdue(3), PartiallyPaid(4)

Appendix B — File Naming Convention

Artifact	Pattern	Example
Commands	`[Action][Entity]Command.cs + Handler.cs`	`CreateCardCommand.cs`
Queries	`[Action][Entity]Query.cs + Handler.cs`	`ListCardsQuery.cs`
Controllers	`[Entity]Controller.cs`	`CardsController.cs`
Repositories (Interface)	`I[Entity]Repository.cs`	`ICardRepository.cs`
Repositories (SQL impl)	`Sql[Entity]Repository.cs`	`SqlCardRepository.cs`
Consumers	`[EventName]Consumer.cs`	`UserDeletedConsumer.cs`
Entities	`[EntityName].cs`	`CreditCard.cs`
DTOs	`[EntityName]Dto.cs`	`CreditCardDto.cs`

Appendix C — Shared Contracts Project Structure

```
shared.contracts/Shared.Contracts/
├── Events/
│   ├── Identity/IdentityEvents.cs        (IUserRegistered, IUserOtpGenerated, IUserDeleted)
│   ├── Card/CardEvents.cs                (ICardAdded)
│   ├── Billing/BillingEvents.cs          (IBillGenerated, IBillOverdueDetected)
│   ├── Payment/PaymentEvents.cs          (IPaymentCompleted, IPaymentFailed, IPaymentReversed, IPaymentOtpGenerated)
│   └── Saga/SagaOrchestrationEvents.cs   (22 saga events)
└── Models/
    └── ApiResponse.cs                    (Shared response wrapper)
```

Document prepared for CredVault Technical Review — CredVault Engineering Team
All diagrams use Mermaid syntax and are renderable in GitHub, GitLab, Notion, and VS Code.